

Problem

Given an integer array of size n , find the majority element. A majority element is an element that appears more than $\lfloor n/2 \rfloor$ times.

If a majority element exists, print it. Otherwise, print -1.

Complete the function `majorityElement` that takes the following parameter:

`int arr[n]`: an array of integers Returns `int`: the majority element if it exists; otherwise -1.

Input Format

The first line contains an integer n , the number of elements in the array. The second line contains n space-separated integers.

Constraints

$1 \leq n \leq 10^5$ $-10^9 \leq arr[i] \leq 10^9$

Output Format

Print the majority element if it exists; otherwise, print -1.

Change Theme

Language

Java 15



```

1  import java.io.*;
2  import java.util.Scanner;
3  class Solution{
4      public static void main(String[] args){
5          Scanner scanner = new Scanner(System.in)
6          int n = scanner.nextInt();
7          int[] arr = new int[n];
8          for(int i=0;i<n;i++){
9              arr[i] = scanner.nextInt();
10         }
11         int candidate = 0;
12         int count = 0;
13         for(int i=0;i<n;i++){
14             if(count==0){
15                 candidate = arr[i];
16                 count = 1;
17             }else if(arr[i]==candidate){
18                 count++;
19             }else{
20                 count--;
21             }
22         }
23         int actualCount = 0;
24         for(int i=0;i<n;i++){
25             if(arr[i]==candidate){
26                 actualCount++;
27             }
28         }

```

Line: 35 Col: 2

Submissions

Leaderboard

Discussions